TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	92.7 ± 1.10
Correlation Stability	91.6 ± 0.66
Distribution Consistency	94.2 ± 1.33
Entropy Preservation	95.1 ± 0.70
Mean Z Score Drift	0.1 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	99.4 ± 0.66
Feature Accuracy	94.9 ± 0.94
Novelty Score	0.8 ± 0.01

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Dataset Audit: heart_failure_clinical_records.csv

1. Dataset Portrait

Attribute	Value
Rows	299
Columns	13
Numeric features	13
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	60.834	60.679	-0.155	0.3%	OK
anaemia	0.431	0.456	+0.025	5.7%	OK
creatinine_phosphokinase	581.839	511.753	-70.086	12.0%	Warn
diabetes	0.418	0.448	+0.030	7.1%	OK
ejection_fraction	38.084	39.598	+1.515	4.0%	OK
high_blood_pressure	0.351	0.414	+0.063	18.0%	Warn
platelets	263358.029	269368.410	+6010.381	2.3%	OK
serum_creatinine	1.394	1.479	+0.085	6.1%	OK
serum_sodium	136.625	137.234	+0.609	0.4%	OK
sex	0.649	0.632	-0.017	2.6%	OK
smoking	0.321	0.314	-0.007	2.3%	OK
time	130.261	126.632	-3.629	2.8%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	92.7 ± 1.10
Correlation Stability (*2)	91.6 ± 0.66
Distribution Consistency (*3)	94.2 ± 1.33
Entropy Preservation (*4)	95.1 ± 0.70
Mean Z Score Drift (*5)	0.1 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	99.4 ± 0.66
Feature Accuracy (*10)	94.9 ± 0.94
Novelty Score (*11)	0.8 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8416666666666666	0.7966666666666666	0.8133333333333332	-0.028333333333333356	
F1	0.7584597087219636	0.6662599534768157	0.7136040592264943	-0.04485564949546926	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.835	0.7816666666666666	0.8116666666666668	-0.023333333333333327	
F1	0.7547418227770806	0.670041668043144	0.7193211838700895	-0.03542063890699111	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7499999999999999	0.77	0.7783333333333332	0.028333333333333332	
F1	0.5667171896628354	0.6188207909321917	0.6272142174913695	0.06049702782853426	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7833333333333334	0.7283333333333333	0.7683333333333333	-0.01499999999999999	
F1	0.6756462612888183	0.6009394783699946	0.6511370082247965	-0.024509253064021852	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4983333333333324	0.495	0.5733333333333335	0.07499999999999998	
F1	0.3315868365248023	0.3160053312327054	0.1037037037037037	-0.22788313282109868	

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8066666666666666	0.7966666666666666	0.805	-0.00166666666666663	
F1	0.6937863401621888	0.6712141362121118	0.6879537638959499	-0.005832576266238942	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8300000000000001	0.7849999999999999	0.8083333333333333	-0.02166666666666664	
F1	0.7541632435801531	0.6725382127496647	0.7191006691786018	-0.03506257440155121	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6133333333333335	0.6183333333333334	0.6233333333333334	0.010000000000000009	
F1	0.007142857142857143	0.08030189853719265	0.1074821641918416	0.10033930704898446	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6333333333333334	0.6333333333333334	0.6333333333333334	0	
F1	N/A	0.0	N/A	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.825	0.7983333333333335	0.8216666666666667	-0.003333333333333435	
F1	0.7417404771442427	0.6872378330677517	0.735990842278561	-0.005749634865681763	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
2	55.0	0.0	7861.0	0.0	38.0	0.0	263358.03	1.1	136.0	1.0
3	65.0	0.0	146.0	0.0	20.0	0.0	162000.0	1.3	129.0	1.0
4	50.0	1.0	111.0	0.0	20.0	0.0	210000.0	1.9	137.0	1.0
5	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
6	90.0	1.0	47.0	0.0	40.0	1.0	204000.0	2.1	132.0	1.0
7	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0
8	60.0	1.0	315.0	1.0	60.0	0.0	454000.0	1.1	131.0	1.0
9	65.0	0.0	157.0	0.0	65.0	0.0	263358.03	1.5	138.0	0.0
10	80.0	1.0	123.0	0.0	35.0	1.0	388000.0	9.4	133.0	1.0
11	75.0	1.0	81.0	0.0	38.0	1.0	368000.0	4.0	131.0	1.0
12	62.0	0.0	231.0	0.0	25.0	1.0	253000.0	0.9	140.0	1.0
13	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
14	50.0	1.0	168.0	0.0	38.0	1.0	276000.0	1.1	137.0	1.0
15	49.0	1.0	80.0	0.0	30.0	1.0	427000.0	1.0	138.0	0.0
16	82.0	1.0	379.0	0.0	50.0	0.0	47000.0	1.3	136.0	1.0
17	87.0	1.0	149.0	0.0	38.0	0.0	262000.0	0.9	140.0	1.0
18	45.0	0.0	582.0	0.0	14.0	0.0	166000.0	0.8	127.0	1.0
19	70.0	1.0	125.0	0.0	25.0	1.0	237000.0	1.0	140.0	0.0
20	48.0	1.0	582.0	1.0	55.0	0.0	87000.0	1.9	121.0	0.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	63.0	0.0	131.0	0.0	14.0	1.0	189000.0	1.83	138.0	1.0
2	40.0	0.0	148.0	1.0	60.0	0.0	422000.0	1.0	140.0	1.0
3	70.0	0.0	2442.0	0.0	35.0	0.0	229000.0	1.0	140.0	1.0
4	61.0	0.0	132.0	0.0	40.0	0.0	360000.0	3.7	134.0	0.0
5	75.0	0.0	910.0	0.0	20.0	1.0	305000.0	1.1	135.0	1.0
6	80.0	0.0	232.0	1.0	30.0	0.0	277000.0	1.1	141.0	1.0
7	45.0	1.0	66.0	1.0	25.0	0.0	233000.0	0.8	135.0	1.0
8	70.0	0.0	161.0	0.0	25.0	0.0	244000.0	1.2	142.0	0.0
9	58.0	1.0	400.0	0.0	40.0	0.0	164000.0	1.0	139.0	0.0
10	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	65.0	0.0	124.0	1.0	60.0	0.0	266000.0	1.2	140.0	1.0
2	51.0	1.0	171.0	0.0	35.0	0.0	179000.0	1.1	144.0	0.0
3	65.0	1.0	132.0	1.0	38.0	1.0	222000.0	1.1	133.0	1.0
4	70.0	0.0	115.0	0.0	30.0	1.0	283000.0	1.0	133.0	1.0
5	60.0	1.0	135.0	1.0	25.0	0.0	203000.0	1.4	143.0	1.0
6	58.0	0.0	190.0	0.0	60.0	1.0	218000.0	0.9	132.0	0.0
7	60.0	0.0	2656.0	1.0	30.0	0.0	305000.0	2.3	137.0	1.0
8	45.0	1.0	130.0	0.0	35.0	0.0	174000.0	0.8	139.0	1.0
9	45.0	0.0	582.0	1.0	55.0	0.0	543000.0	1.0	132.0	0.0
10	65.0	1.0	59.0	1.0	60.0	0.0	172000.0	0.9	137.0	0.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	53.0	0.0	80.0	0.0	17.0	1.0	220000.0	5.8	143.0	1.0
2	53.0	1.0	59.0	1.0	40.0	1.0	218000.0	1.1	137.0	1.0
3	58.0	0.0	1199.0	0.0	20.0	0.0	263000.0	0.8	142.0	0.0
4	49.0	1.0	1876.0	1.0	60.0	1.0	266000.0	0.8	142.0	0.0
5	42.0	0.0	395.0	0.0	25.0	0.0	255000.0	1.7	137.0	1.0
6	58.0	1.0	707.0	0.0	45.0	0.0	279000.0	9.0	141.0	1.0
7	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0
8	79.0	1.0	55.0	0.0	50.0	1.0	172000.0	1.8	133.0	1.0
9	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
10	70.0	0.0	92.0	0.0	60.0	1.0	317000.0	0.8	140.0	0.0

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B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	60.0	0.0	70.0	1.0	38.0	0.0	213000.0	3.5	134.0	0.0
2	50.0	0.0	157.0	1.0	20.0	1.0	237000.0	1.5	142.0	1.0
3	75.0	0.0	250.0	0.0	38.0	0.0	244000.0	5.0	135.0	0.0
4	70.0	1.0	582.0	1.0	60.0	0.0	324000.0	1.1	141.0	0.0
5	45.0	0.0	47.0	0.0	50.0	0.0	263358.0	1.83	136.0	1.0
6	45.0	0.0	4540.0	1.0	40.0	0.0	281000.0	1.0	140.0	0.0
7	48.0	1.0	131.0	1.0	30.0	1.0	244000.0	1.6	130.0	0.0
8	95.0	1.0	371.0	0.0	30.0	0.0	461000.0	2.0	132.0	1.0
9	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
10	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	50.0	0.0	577.0	0.0	25.0	0.0	282000.0	0.7	136.0	0.0
2	53.0	0.0	75.0	1.0	38.0	0.0	277000.0	1.18	138.0	0.0
3	40.0	1.0	835.0	1.0	50.0	0.0	203000.0	1.3	138.0	1.0
4	81.0	1.0	84.0	0.0	25.0	0.0	274000.0	1.1	136.0	0.0
5	52.0	0.0	1380.0	1.0	25.0	1.0	274000.0	1.5	133.0	0.0
6	85.0	1.0	298.0	1.0	38.0	0.0	221000.0	1.8	136.0	0.0
7	78.0	0.0	224.0	0.0	50.0	0.0	481000.0	1.4	138.0	1.0
8	70.0	0.0	582.0	0.0	20.0	1.0	263358.03	1.83	134.0	1.0
9	61.0	0.0	248.0	0.0	30.0	1.0	267000.0	0.7	136.0	1.0
10	40.0	0.0	478.0	1.0	30.0	0.0	303000.0	0.9	136.0	1.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	68.0	1.0	25.0	1.0	226000.0	1.0	134.0	0.0
2	40.0	1.0	96.0	0.0	38.0	1.0	194000.0	1.1	134.0	1.0
3	68.0	0.0	212.0	0.0	60.0	0.0	277000.0	1.3	140.0	0.0
4	45.0	0.0	143.0	1.0	50.0	1.0	451000.0	1.18	142.0	0.0
5	72.0	1.0	121.0	0.0	30.0	0.0	226000.0	1.3	139.0	1.0
6	55.0	0.0	1202.0	1.0	35.0	0.0	189000.0	1.0	142.0	0.0
7	81.0	0.0	4540.0	0.0	35.0	0.0	231000.0	1.18	137.0	1.0
8	70.0	0.0	2695.0	1.0	40.0	0.0	241000.0	1.0	137.0	1.0
9	70.0	0.0	835.0	0.0	35.0	1.0	305000.0	0.8	133.0	0.0
10	80.0	0.0	776.0	1.0	38.0	1.0	192000.0	1.3	135.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	85.0	0.0	1846.0	0.0	15.0	0.0	51000.0	1.0	136.0	0.0
2	72.0	0.0	315.0	0.0	30.0	0.0	302000.0	0.9	136.0	0.0
3	42.0	1.0	897.0	1.0	60.0	0.0	87000.0	0.7	136.0	1.0
4	60.0	0.0	582.0	1.0	38.0	0.0	271000.0	1.2	136.0	1.0
5	55.0	0.0	298.0	1.0	38.0	0.0	141000.0	0.7	138.0	1.0
6	42.0	0.0	64.0	0.0	25.0	0.0	226000.0	1.2	138.0	1.0
7	75.0	0.0	582.0	1.0	30.0	1.0	263358.03	1.83	134.0	0.0
8	53.0	0.0	207.0	1.0	40.0	0.0	223000.0	1.2	130.0	0.0
9	60.0	1.0	2281.0	1.0	40.0	0.0	283000.0	1.0	141.0	0.0
10	59.0	1.0	280.0	1.0	25.0	1.0	302000.0	1.0	141.0	0.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	1.0	1688.0	0.0	35.0	0.0	533000.0	0.9	136.0	0.0
2	55.0	0.0	47.0	0.0	30.0	0.0	301000.0	1.0	137.0	1.0
3	55.0	0.0	1419.0	1.0	30.0	1.0	276000.0	0.9	136.0	0.0
4	55.0	0.0	47.0	0.0	35.0	0.0	204000.0	0.6	139.0	1.0
5	90.0	0.0	68.0	1.0	35.0	0.0	228000.0	1.2	133.0	1.0
6	45.0	0.0	59.0	0.0	80.0	0.0	263358.0	1.83	140.0	1.0
7	65.0	0.0	224.0	1.0	50.0	0.0	149000.0	1.3	137.0	1.0
8	80.0	0.0	898.0	0.0	25.0	0.0	149000.0	1.1	144.0	1.0
9	66.0	1.0	72.0	0.0	40.0	1.0	242000.0	1.2	134.0	1.0
10	75.0	1.0	582.0	0.0	30.0	0.0	225000.0	1.83	134.0	1.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	60.0	0.0	47.0	0.0	35.0	0.0	406000.0	4.0	124.0	1.0
2	58.0	1.0	176.0	0.0	40.0	0.0	305000.0	1.5	140.0	0.0
3	82.0	0.0	315.0	0.0	55.0	0.0	317000.0	0.7	130.0	1.0
4	63.0	0.0	446.0	0.0	50.0	0.0	218000.0	1.3	139.0	1.0
5	60.0	0.0	157.0	0.0	40.0	0.0	248000.0	3.2	132.0	0.0
6	65.0	0.0	86.0	0.0	60.0	1.0	219000.0	1.0	145.0	1.0
7	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0
8	60.0	0.0	68.0	0.0	20.0	0.0	119000.0	2.9	127.0	1.0
9	70.0	0.0	97.0	0.0	60.0	1.0	220000.0	0.9	138.0	1.0
10	53.0	1.0	1808.0	0.0	60.0	1.0	249000.0	0.7	138.0	1.0

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	4540.0	0.0	38.0	0.0	172000.0	0.8	140.0	1.0
2	45.0	0.0	159.0	1.0	25.0	0.0	232000.0	1.7	140.0	0.0
3	73.0	1.0	129.0	1.0	35.0	1.0	365000.0	1.0	134.0	0.0
4	52.0	0.0	135.0	0.0	30.0	0.0	362000.0	1.2	134.0	1.0
5	85.0	1.0	47.0	1.0	38.0	0.0	226000.0	1.4	142.0	0.0
6	80.0	1.0	47.0	1.0	60.0	1.0	263358.0	1.0	138.0	1.0
7	60.0	0.0	897.0	1.0	45.0	0.0	297000.0	1.0	133.0	1.0
8	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
9	45.0	1.0	1876.0	1.0	35.0	0.0	226000.0	0.9	138.0	1.0
10	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0